



Department of Education

STEM Education
National Schools of Excellence
Biology Syllabus
Grade 12
Term 2

<u>DOMAIN</u>	<u>Phylum</u>
SCIENCE	Biology
Technology	
Engineering	
Mathematics	

Issued free to schools by the Department of Education

Published in 2021 by the Department of Education, Papua New Guinea

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ISBN

Acknowledgements

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

This document was developed with support from the General Education Services Division of the Department of Education.

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SECRETARY'S MESSAGE

This Biology Syllabus is to be used exclusively by teachers of Biology to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Biology Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Biology before undertaking the advanced concepts introduced in this syllabus.

The syllabus conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Biology as a fundamental field of science.

This STEM Biology Syllabus for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Biology.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Biology is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Biology to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this syllabus is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this syllabus as the official curriculum for Biology to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Biology is the study of living organisms and how they interact at multiple levels to sustain life. The basic building block of life is considered to be the cell. The hereditary units are the genetic sequences, and these are encased within the cells. This genetic information ensures the hereditary traits necessary for survival of the species is passed on from parents to progenies. Biology is divided into different fields of study. For instance, botany is the study of plants whilst zoology involves the study of animals. Micro-organisms are studied in microbiology. Recent fields that emerged in the last few decades include molecular biology, genomics, and so on.

Since Biology is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Biology. Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

The **Grade 12** Biology Syllabus entails **six (6)** Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and **four (4)** topics. Over time different Modules will be introduced.

THEMA	MODULE	TOPICS
BIOTECHNOLOGY	12.1: Bioinformatics	1. Transcription and Translation 2. DNA Sequence Data Analysis 3. Bioinformatics 4. DNA Extraction and Sequencing
HEALTH AND MEDICINE	12.1: Health and Medical Biology	1. Disease – Disruption of Homeostasis 2. Systemic Response to Disease or Injury 3. Biology of Hemostasis 4. Basics of Disease Diagnosis 5. Surgery – Dissection of Animals
ENERGY	12.1: Bioenergy	1. Energy Conversion by Photosynthetic Organisms 2. Production of Fuel Alcohol from Cellulosic Biomass 3. Biomass and Biofuels 4. Biogas Production
BATTERY TECHNOLOGY	12.1: Bio-Batteries	1. Introduction to Biobatteries 2. Paper batteries 3. Sugar Batteries 4. Bioelectrochemical Reactor 5. The Moses Experiment – Burning Bush
ARTIFICIAL INTELLIGENCE	12.1: Neuroscience	1. General Design of the Nervous System

		2. Sensory Receptors and Neuronal Circuits for Processing Information 3. Sense – The Eye and Vision 4. Cerebral Cortex, Intellectual Functions of Brain, Learning, and Memory 5. Behavioral and Motivational Mechanisms of the Brain – The limbic System
ROBOTICS	12.1: Biological Inspirations to Robotics	1. Motor and Integrative Neurophysiology 2. Sensory Systems 3. Biologically Inspired Robots 4. Cyborg

Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Syllabus are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary. The STEM Bridging Syllabus is a bridge across the divide that Grade 11 Students enrolling for STEM Education must cross.

In the Biology Bridging Syllabus students were introduced to the basic concepts and knowledge of Biology in order for them to develop an appreciation of the discipline in a broader context. In this syllabus students are introduced to advanced concepts in Biology as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. These syllabi will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development new nomenclature were introduced. These are explained below:

Term	Explanation
Domain	Each STEM area is defined as <i>domain</i> (e.g. Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as “subject” (e.g. Biology, Physics, Chemistry, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> .
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domains

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM BIOLOGY CONTENT OVERVIEW – TERM 2

Table 1.1 Sequencing & Benchmarking STEM Chemistry Content Standards

Setting STEM Chemistry Content Standards for Progressive Learning & Assessment			
Thema	Broad Content Standard		
Energy			
Module 12.1 Bioenergy	12.1a: Appreciate the natural energy sources and how to harness the full potential of renewable energy within the country (PNG). 12.1b: Identify the comparative advantage of renewable energy sources over non-renewable energy sources. 12.1c: Demonstrate an Understanding and appreciation for the emerging sustainable renewable technologies. 12.1d: Analyze the market, industry and evaluate the policy trends of renewable energy. 12.1e: Conceptualize the geopolitics of energy within the country, Papua New Guinea.		
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1. Energy Conversion by Photosynthetic Organisms	12.1.1a: Describe the photosynthetic capture of solar energy 12.1.1b: Photosynthesis mechanisms 12.1.1c: Hydrogen production through solar energy conversions	Student's ability to master the knowledge and skills of: • Photosynthesis conversion of energy • Production of biofuels • Production of Biogas	Quiz + Written Test + Assignment + Lab Practical
2. Production of Fuel Alcohol from Cellulosic Biomass	12.1.2a: Utilize microorganisms to produce Cellulase 12.1.2b: Convert Cellulosic Waste Materials via Saccharification 12.1.2c: Use Immobilize Yeast Cells to ferment alcohol 12.1.2d: Design and propose a small-scale plant that produces biofuels from using Cellulase wastes and immobilize yeast; and how this can be implement in Papua New Guinea.		
3. Biomass and Biofuels	12.1.3a: Explain what Biomass feedstock is 12.1.3b: Discuss the dictated energy crops that are available in Papua New Guinea		

4. Biogas Production	12.1.4a: Experiment how to produce biofuel or biogas at home. 12.1.4b: Appreciate and promote sustainable energy production.		
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Thema Battery Technology	Broad Content Standard		
Module 12.1 Bio-Batteries	12.1a: Discuss and describe the paper batteries derived from lignin and cellulose in plants. 12.1b: Discuss and describe the sugar batteries from glucose sources in animals (especially humans) and plants also. 12.1c: Describe the Bioelectrochemical reactors from the microbial sources of batteries and their applications in water treatment. 12.1d: Appreciate the evolution of batteries from initial studies on frogs and current trends in battery technology. 12.1e: Appreciate the sustainable development of batteries from biological sources.		
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1. Introduction to Biobatteries	12.1.1a: Define Biobattery 12.1.1b: Explain how biobatteries work 12.1.1c: Discuss the advantages and disadvantages of using Biobatteries 12.1.1d: List and discuss some examples of how biobatteries can change energy storage 12.1.1e: Explain how Bacteria can act as an energy storage or battery 12.1.1f: Research and present an idea on a particular bacteria's potential in acting as a energy storage or battery.	Student's ability to master the knowledge and skills of: • Biobatteries • Bioelectrochemical Reactor • Plants electrical signals	Quiz + Written Test + Assignment + Lab Practical
2. Paper Batteries	12.1.2a: Evaluate the advantages and disadvantages of paper batteries.		

	12.1.2b: Describe the potential applications of the paper batteries. 12.1.2c: Attempt in constructing a paper battery		
3. Sugar Battery	12.1.3a: Identify and describe the potential benefits, drawbacks, design, and improvements of sugar batteries. 12.1.3b: Attempt to construct a sugar battery		
4. Bioelectrochemical Reactor	12.1.4: Appreciate the history and describe the working principles and applications of a Bioelectrochemical reactor in water treatment.		
5. The Moses Experiment: 'Burning Bush'	12.1.5a: Describe the photosystems in plants, the electron transport chain and the generation of energy. 12.1.5b: Appreciate the concept of energy storage and generation in plants that might fuel the next generation.		

Table 1.2 Mapping Content Progressions by Modules

Thema/Module	Broad Content Standard	12.1	12.1
Energy/ Module 12.1	12.1a: Appreciate the natural energy sources and how to harness the full potential of renewable energy within the country (PNG). 12.1b: Identify the comparative advantage of renewable energy sources over non-renewable energy sources. 12.1c: Demonstrate an Understanding and appreciation for the emerging sustainable renewable technologies. 12.1d: Analyze the market, industry and evaluate the policy trends of renewable energy. 12.1e: Conceptualize the geopolitics of energy within the country, Papua New Guinea.	✓	
Battery Technology/ Module 12.1	12.1a: Discuss and describe the paper batteries derived from lignin and cellulose in plants. 12.1b: Discuss and describe the sugar batteries from glucose sources in animals (especially humans) and plants also. 12.1c: Describe the Bioelectrochemical reactors from the microbial sources of batteries and their applications in water treatment. 12.1d: Appreciate the evolution of batteries from initial studies on frogs and current trends in battery technology. 12.1e: Appreciate the sustainable development of batteries from biological sources.		✓

THEMA: ENERGY

Introduction

The ability to obtain (relatively) affordable energy has become essential to the operation of modern economies. However, substantial vulnerabilities have emerged as a consequence of the uneven distribution of energy supply among countries. International energy relations have facilitated in the globalization of the world, culminating in both energy security and vulnerability. In contrast to other energy sources, which are concentrated in a small number of nations, renewable resources and major prospects for energy efficiency and transitions exist across a large geographical area. Rapid deployment of renewable energy and energy efficiency, as well as technical energy source diversification, would result in major energy security and economic benefits.

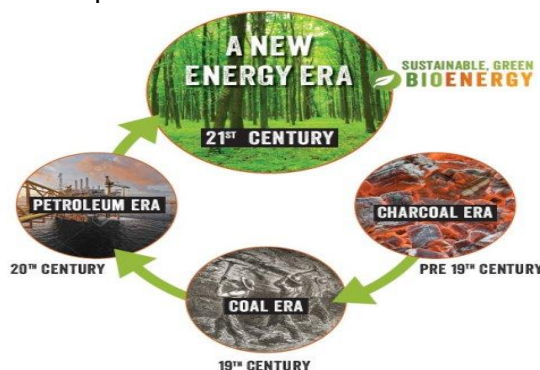
The global energy supply chain and consumption is key to the economic activities that govern a nation. Energy supply is mostly reliant on non-renewable sources of energy as in fossil fuels or hydrocarbons (coal, oil and natural gas) with nuclear fuels and marginally from renewable sources (hydropower, solar, wind, geothermal). Because of the increased reliance on carbon-emitting energy sources such as fossil fuels and carbon-emitting renewables such as biomass during the twentieth century, the energy industry has been a major contributor to pollution and the economy's environmental repercussions. Fossil fuels were the primary source of energy generation in most regions of the world until recently, and they constitute a significant contributor to global warming and pollution. Many economies are investing in renewable and sustainable energy as part of human response to global warming.

Therefore, much of the discussions below would be concentrated on sustainable renewable sources of energy and their emerging technologies.

MODULE 12.1: BIOENERGY

4-5 Weeks

Bioenergy is energy derived from biomass or biofuel. Biomass is any organic material that has absorbed sunlight and stored it as chemical energy. Wood, energy crops, and wastes from woods, yards, or farms are a few examples. The term "biofuel" is typically reserved for liquid or gaseous fuels utilized in transportation.



(Link: www.bioenergy.org.nz)

Linking Modules

Prerequisites:

STEM Biology Grade 11 Bridging:

- Module 6: Ecology

STEM Biology Grade 11 Term 2:

- Thema: Biodiversity
- Thema: Biotechnology

STEM Biology Grade 11 Term 3:

- Thema: Energy

STEM Biology Grade 11 Term 4:

- Thema: Earth Science

Broad Content Standard

12.1a: Appreciate the natural energy sources and how to harness the full potential of renewable energy within the country (PNG).

12.1b: Identify the comparative advantage of renewable energy sources over non-renewable energy sources.

12.1c: Demonstrate an Understanding and appreciation for the emerging sustainable renewable technologies.

12.1d: Analyze the market, industry and evaluate the policy trends of renewable energy.

12.1e: Conceptualize the geopolitics of energy within the country, Papua New Guinea.

Content Benchmark

Student's ability to master the knowledge and skills of:

- Photosynthesis conversion of energy
- Production of biofuels
- Production of Biogas

Topic 1: Energy conversion by photosynthetic organisms

Performance Indicator

Students are expected to:

12.1.1a: Describe the photosynthetic capture of solar energy

12.1.1b: Photosynthesis mechanisms

12.1.1c: Hydrogen production through solar energy conversions

Content Expansion

Body of Knowledge for Understanding

- Photosynthetic Capture of Solar Energy
 - Solar Energy
 - Why is Biotechnology now Applied in Energy Technology?
- Photosynthesis Mechanisms
 - Plant Photosynthesis
 - Bacterial Photosynthesis
- Hydrogen Production through solar Energy Conversions
 - Cyanobacterial Hydrogen Production (Plant-Type Photosynthesis)
 - Bacterial Hydrogen Production (Bacterial-Type Photosynthesis)
 - Use of Photosynthesized Proteins in Photoelectric Conversion Elements

Set of Proficiency Skills

- Analytical problem solving
- Research Skills
- Numerical Skills
- Written and Oral Communication skills
- Observational Skills
- Teamwork and interpersonal skills

Traits of Attitudes and Values

- Patience
- Determination
- Flexibility
- A logical and independent mind
- Meticulous attention to detail and accuracy

Topic 2: Biomass

Performance Indicators

Students are expected to:

12.1.2a: Analyze the essence of biomass as a renewable energy source

12.1.2b: Describe the primary sources of biomass

12.1.2c: Describe the secondary sources of biomass

12.1.2d: evaluate the different processes available for conversion of biomass.

Content Expansion

Body of Knowledge for Understanding

- Introduction
- Primary Biomass
- Secondary Biomass
- Biomass Conversion
 - Thermal Conversion
 - Chemical Conversion
 - Biological Conversion

Traits of Values

- Appreciate the field of biotechnology and engineering in producing alternative energy .

Set of Proficiency Skills

- Derive potentials for harnessing biomass and conversion into energy

Topic 3: Biofuels

Performance Indicators

Students are expected to:

12.1.3a: Explain what Biomass feedstock is

12.1.3b: Discuss the dictated energy crops that are available in Papua New Guinea

Content Expansion

Body of Knowledge for Understanding

- First Generation Biofuels
- Second Generation Biofuels
- Third Generation Biofuels
- Fourth Generation Biofuels

Traits of Values

- Appreciate and value the rich energy resources in Papua New Guinea

Set of Proficiency Skills

- Pre-plan alternative energy source systems that can power Papua New Guinea in the long run.

Topic 4: Biogas Production

Performance Indicators

Students are expected to:

12.1.4a: Experiment how to produce biofuel or biogas at home.

12.1.4b: Appreciate and promote sustainable energy production.

Content Expansion

Body of Knowledge for Understanding

- Experiment: Production of Biogas
 - Biogas Basics
 - Five (5) steps to Making Homemade Biogas

Traits of Values

- Appreciate the process of producing Biogas

Set of Proficiency Skills

- Able to Produce biogas

THEMA: BATTERY TECHNOLOGY

Introduction

Batteries have been around for a long time and have played a significant role in shaping and improving humanity's massive leap into the modern era. They've evolved into a vital and practical gadget for recharging batteries. Batteries, in essence, store chemical energy that is converted into electric energy and used by other devices. The batteries are tiny chemical reactors that, as a final reaction, produce energetic electrons that pass through and power connected electronics. Batteries have become so commonplace in modern life that it's difficult to fathom life without them. Batteries, on the other hand, have a short life span and hence cannot power equipment; they are constructed of potentially hazardous chemicals; they are non-renewable; and they are not environmentally friendly. Previously, this battery theme discussed the challenges and, more significantly, looked at alternative chemical batteries that can offer the desired results. Here, this module considers the necessity to research nature in order to embrace its knowledge and develop bio-batteries.

MODULE 12.1: BIO-BATTERIES

4-5 Weeks

A bio-battery is an energy storage device that runs on organic substances, most commonly glucose, such as the glucose found in human blood. Multiple electrons and protons are released when enzymes in human bodies break down glucose. Bio-batteries may directly obtain energy from glucose by employing enzymes to break it down. The energy is then stored in these batteries and used later. Plants and many animals use energy from metabolic reactions in nearly identical ways.

Several research teams and engineers are attempting to further progress the development of these batteries, even if they are still being tested before being commercially supplied.

Linking Modules

Prerequisites:

STEM Biology Grade 11 Bridging:

- Module 2: Cells
- Module 3: Metabolism

STEM Biology Grade 11 Term 3:

- Thema: Battery Technology

Broad Content Standard

12.1a: Discuss and describe the paper batteries derived from lignin and cellulose in plants.

12.1b: Discuss and describe the sugar batteries from glucose sources in animals (especially humans) and plants also.

12.1c: Describe the Bioelectrochemical reactors from the microbial sources of batteries and their applications in water treatment.

12.1d: Appreciate the evolution of batteries from initial studies on frogs and current trends in battery technology.

12.1e: Appreciate the sustainable development of batteries from biological sources.

Content Benchmark

Student's ability to master the knowledge and skills of:

- Biobatteries
- Bioelectrochemical Reactor
- Plants electrical signals

Topic 1: Introduction to Biobatteries

Performance Indicators

Students are expected to:

12.1.1a: Define Biobattery

12.1.1b: Explain how biobatteries work

12.1.1c: Discuss the advantages and disadvantages of using Biobatteries

12.1.1d: List and discuss some examples of how biobatteries can change energy storage

12.1.1e: Explain how Bacteria can act as an energy storage or battery

12.1.1f: Research and present an idea on a particular bacteria's potential in acting as a energy storage or battery.

Content Expansion

Body of Knowledge for Understanding

- What is a Biobattery?
- How Biobatteries work
- Advantages of Biobatteries
- Disadvantage of Biobatteries
- Give examples of Applications of Biobatteries
- Bacteria as an energy storage or Battery

Traits of Values

- Appreciate the role of battery and the advancement of battery technology through biological systems

Set of Proficiency Skills

- Research skills on different bacteria

Topic 2: Paper Batteries

Performance Indicators

Students are expected to:

12.1.2a: Evaluate the advantages and disadvantages of paper batteries.

12.1.2b: Describe the potential applications of the paper batteries.

12.1.2c: Attempt in constructing a paper battery

Content Expansion

Body of Knowledge for Understanding

- What is Paper Battery?
- Advantages
- Disadvantages
- Electrolyte for Paper Battery
- Potential Applications
 - Electrochemical Batteries
 - Li-ion Batteries
 - Biofuel Cells
 - Supercapacitors
 - Nanogenerators

Traits of Values

- Appreciate the role of battery and the advancement of battery technology through biological systems

Set of Proficiency Skills

- Able to construct a paper battery

Topic 3: Sugar Battery

Performance Indicators

Students are expected to:

12.1.3a: Identify and describe the potential benefits, drawbacks, design and improvements of sugar batteries.

12.1.3b: Attempt to construct a sugar battery

Content Expansion

Body of Knowledge for Understanding

- Sugar Battery
- History
- Potential benefits
- Safety
- Availability of Fuel
- Safety
- Availability of Fuel
- Environmental Friendliness
- High Energy Density
- Drawbacks
 - Relatively Low Voltage
 - Water Requirement
 - Design
 - Anode
 - Synthetic Pathway
- Improvements
 - Non-immobilized Enzymes
 - Thermo Enzymes
 - Synthetic Catabolic Pathway

Traits of Values

- Appreciate the role of battery and the advancement of battery technology through biological systems

Set of Proficiency Skills

- Able to construct a sugar battery

Topic 4: Bioelectrochemical Reactor

Performance Indicators

Students are expected to:

12.1.4: Appreciate the history and describe the working principles and applications of a Bioelectrochemical reactor in water treatment.

Content Expansion

Body of Knowledge for Understanding

- Bioelectrochemical Reactor: What is it?
- Principles
- History
- Applications
 - Water Treatment
 - Shewanella Oneidensis
 - Name
 - Qualities
 - Metal Reduction
 - Chemical modification
 - Pellicle Formation
 - Nanotechnology
 - Wastewater Treatment
 - Genome
- Bacterial Nanowires
 - Physiology
 - History
 - Implications and Potential Applications
 - Biological Implications
 - Bioenergy Applications in Microbial Fuel Cells
 - Other Significant Applications

Traits of Values

- Appreciate Bioelectrochemical reactors

Set of Proficiency Skills

- Explain the principles that govern the mechanism of Bioelectrochemical reactor

Topic 5: The Moses Experiment: 'Burning Bush'

Performance Indicators

Students are expected to:

12.1.5a: Describe the photosystems in plants, the electron transport chain and the generation of energy.

12.1.5b: Appreciate the concept of energy storage and generation in plants that might fuel the next generation.

Content Expansion

Body of Knowledge for Understanding

- Introduction
- Experimental details
 - Investigation of Type of Electrodes and Energy Sources
 - Optimization of the Harvesting System
 - Investigation on the Potential Application
- Results and Discussions
 - Selection of Electrode pair and Energy Source
 - Power Output Optimization
 - Test on Some Practical Application
- Conclusion

Traits of Values

- Appreciate the structure of Plants and their potential to be a source of light (energy)

Set of Proficiency Skills

- Investigative skills